WORKED SOLUTION

STEP III 1988, Question 14

Mechanics • Motion on a Smooth Wire

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Read the question before you go on. It is in the STEP Support Programme's question database:

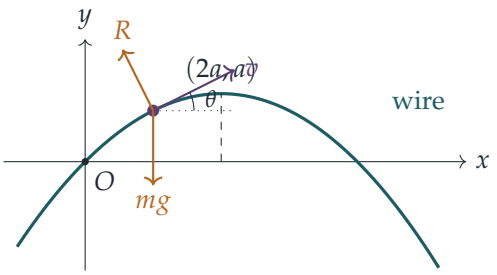
step.maths.org/questions/88-s3-q14

The key idea

A bead that feels no force from the wire is simply a projectile, moving under gravity alone. So part (i) asks: can a free projectile, launched along the wire, follow the wire's parabola exactly? Match the two parabolas. For part (ii), the wire is smooth, so energy is conserved. The work is in writing the speed and the height in terms of the slope angle θ, using $\tan\theta = \mathrm{d}y/\mathrm{d}x$.

Solution

We model the bead as a particle of mass m on a smooth fixed wire. The only forces are its weight mg, acting downwards, and the normal reaction R of the wire, which is perpendicular to the wire.



Two facts about the wire are used throughout. At a general point,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = 1 - \frac{x}{2a},$$

so at the origin the slope is 1 and the bead starts at 45° above the x-axis. Since the wire is a parabola opening downwards, its highest point is at $x = 2a$, $y = a$.

Part (i)

Suppose $R = 0$ throughout the motion. Then gravity is the only force, so the bead moves exactly as a projectile with initial speed V at $45°$. Its horizontal velocity is constant at $V/\sqrt{2}$, so $x = Vt/\sqrt{2}$. Also

$$y = \frac{V}{\sqrt{2}}\,t - \tfrac{1}{2}gt^2.$$

Eliminating $t = \sqrt{2}\,x/V$ gives the trajectory

$$y = x - \frac{gx^2}{V^2}.$$

This is the path of the bead only if it is the wire, $y = x - x^2/(4a)$, for every x the bead reaches. That needs

$$\frac{g}{V^2} = \frac{1}{4a}, \qquad \text{that is,} \qquad V = 2\sqrt{ag}.$$

Conversely, take $V = 2\sqrt{ag}$. A free particle projected from O at $45°$ with this speed follows the curve $y = x - x^2/(4a)$, which is the wire itself. So it moves along the wire with $R = 0$ at every instant, and this motion satisfies the bead's equation of motion with zero reaction. Because $x = Vt/\sqrt{2}$ increases steadily, the bead passes over the top at $(2a, a)$ and continues down the far side for all time. Throughout, it exerts no pressure on the wire, as required, and the suitable speed is $V = 2\sqrt{ag}$.

Ask yourself How can a bead on a wire move with no reaction at all? Doesn't the wire have to hold it on?

The wire only has to push when gravity alone would carry the bead off the curve. For any other speed it would. A larger V gives a flatter projectile parabola, so the wire must pull the bead down. A smaller V gives a more sharply curved one, so the wire must hold it up. At exactly $V = 2\sqrt{ag}$, gravity already supplies the acceleration the curve requires, and the wire has nothing to do.

Part (ii)

Now V is general. The velocity is along the tangent to the wire, so

$$\tan\theta = \frac{\mathrm{d}y}{\mathrm{d}x} = 1 - \frac{x}{2a}, \qquad \text{so} \qquad x = 2a(1 - \tan\theta).$$

(If the bead slides back, its velocity points the opposite way along the tangent and its angle changes by π. Then $\tan\theta$ and $\cos^6\theta$ are unchanged, so nothing below is affected.)

Speed in terms of θ. Differentiating $x = 2a(1 - \tan\theta)$ with respect to t gives

$$\dot{x} = -2a\sec^2\theta\,\dot{\theta}.$$

Since $\dot{y} = \tan\theta\,\dot{x}$, the speed v satisfies

$$v^2 = \dot{x}^2 + \dot{y}^2 = \dot{x}^2(1 + \tan^2\theta) = \dot{x}^2\sec^2\theta = \frac{4a^2\dot{\theta}^2}{\cos^6\theta}.$$

Height in terms of θ. Writing $T = \tan\theta$, so that $x = 2a(1-T)$,

$$\begin{aligned} y = x - \frac{x^2}{4a} &= 2a(1-T) - a(1-T)^2 \\ &= a(1-T)\big(2-(1-T)\big) \\ &= a(1-T)(1+T) = a(1-\tan^2\theta). \end{aligned}$$

Energy. The wire is smooth and R is perpendicular to the velocity, so R does no work and mechanical energy is conserved. The bead starts at O with speed V, so

$$\tfrac{1}{2}mv^2 + mgy = \tfrac{1}{2}mV^2, \qquad \text{that is,} \qquad v^2 + 2gy = V^2.$$

Substituting the expressions for v^2 and y gives

$$\frac{4a^2\dot\theta^2}{\cos^6\theta} + 2ga(1-\tan^2\theta) = V^2,$$

which is the required result.

Checks

- At $t = 0$ we have $\theta = 45^\circ$, so $\tan^2\theta = 1$ and the second term vanishes. The first term is then $v^2 = V^2$, as it should be.
- With $V^2 = 4ag$ from part (i), $\dot x = V/\sqrt{2}$ is constant, so $\dot\theta = -\dot x\cos^2\theta/(2a)$. The first term becomes $2ag\sec^2\theta = 2ag(1+\tan^2\theta)$. Adding $2ga(1-\tan^2\theta)$ gives $4ag = V^2$. The two parts agree.
- Numerically, we stepped the bead's equation of motion along the wire with a Runge–Kutta method. With $a = 1$, $g = 9.8$ and $V = 2\sqrt{9.8}$, the computed normal reaction stayed below 10^{-15} (per unit mass) all the way to $x \approx 13$. With V increased by 30%, it was about 4.8 at the start. For random values of V, including some where the bead slides back, the identity in (ii) held to a relative error of about 10^{-13}.

Common slips

- Showing only that $R = 0$ at the instant of projection. The question wants a motion with no reaction *throughout*, which is why we match the whole trajectory to the wire.
- Forgetting the chain rule factor $\sec^2\theta$ when differentiating $\tan\theta$. This leaves $\cos^2\theta$ instead of $\cos^6\theta$ in the first term.

Beyond STEP

Part (ii) is a first integral of the motion: energy conservation reduces a second-order equation to a first-order one in a single coordinate, here θ. In a first-year course this becomes the Lagrangian approach to constrained motion, where you pick one

coordinate that describes the constraint and never need to calculate the reaction at all. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see constrained motion and energy methods treated carefully.

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